Homework 7. ARMA

You can get up to 10 points for this homework.

Question 1. The Table 1 on the next page presents the CO_2 readings from Mauna Loa. Develop an appropriate ARMA model and a procedure for forecasting for these data.

For the following tasks use Python and submit one file incorporating all your answers, named as "H7-YourFirstName-YourLastName.ipynb".

- $\boxed{\text{A}}$ Create a .csv file out of the given data and put it online (you will use a link to it in your program).
- $\boxed{\text{B}}$ Plot the data.
- $\boxed{\text{C}}$ Run the ADF test and check if the process is stationary. If not, apply differencing.
- $\boxed{\text{D}}$ Separate your series into a train set and a test set.
- $\boxed{\text{E}}$ Define a range of values for p and q, and generate combinations of orders (p, q).
- $\boxed{\text{F}}$ Fit all unique ARMA(p, q) models, and select the one with the lowest AIC.
- $\boxed{\text{G}}$ Select the best model according to the AIC, and store the residuals in a variable.
- $\boxed{\text{H}}$ Perform a qualitative analysis of the residuals with the plot_diagnostics method. Does the Q-Q plot show a straight line that lies on $y = x$?
- $\boxed{\text{I}}$ Perform a quantitative analysis of the residuals by applying the Ljung-Box test on the first 10 lags. Are all returned p-values above 0.05? Are the residuals correlated or not?
- $\boxed{\text{J}}$ Make predictions using the selected ARMA(p, q) model, the mean method, and the last known value method.
- $\boxed{\text{K}}$ Plot your forecasts.
- $\boxed{\text{L}}$ Evaluate each method's performance using the MSE. Which method performed best?
- $\boxed{\text{M}}$ Reverse differencing, plot the best forecast and calculate the MAE.

Table 1. Atmospheric CO_2 Concentrations at Mauna Loa Observatory

Year	Average CO_2 Concentration (ppmv)
1959	316.00
1960	316.91
1961	317.63
1962	318.46
1963	319.02
1964	319.52
1965	320.09
1966	321.34
1967	322.13
1968	323.11
1969	324.60
1970	325.65
1971	326.32
1972	327.52
1973	329.61
1974	330.29
1975	331.16
1976	332.18
1977	333.88
1978	335.52
1979	336.89
1980	338.67
1981	339.95
1982	341.09
1983	342.75
1984	344.44
1985	345.86
1986	347.14
1987	348.99
1988	351.44
1989	352.94
1990	354.19
1991	355.62
1992	356.36
1993	357.10
1994	358.86
1995	360.90
1996	362.58
1997	363.84
1998	366.58
1999	368.30
2000	369.47
2001	371.03
2002	373.07
2003	375.61